

ENV 4120

Air Pollution and Hazardous Waste Control

Today's Topics

- Acid deposition and stratospheric ozone depletion
- Expectation
 - Understand types of deposition; pH and pH of natural rain water; important chemicals that lead to acid rain; and trend of rain pH in the US
 - Understand good and bad ozone; vertical and horizontal structure, and function of ozone layer; what is ozone hole and DU; and the role of CFC and PSC in ozone hole

Acid Deposition

Deposition

- Transfer of pollutants (and pollutant products) from the atmosphere to the earth's surface
- Have effects at many scales
 - Acidification of lakes (and soils)
 - Material degradation
 - Toxic metals deposition (e.g. Hg)
 - Persistent organic pollutants
 - Nitrogen input to ecosystems

Dry and wet deposition

- Dry deposition: Direct deposition to surfaces
- Wet deposition: Precipitation involved deposition
- Acid deposition: Creation and transport of acid constituents from atmosphere to earth surface
 - Can be both dry and wet deposition
 - Chemical reactions usually involved in both gaseous and aqueous phase

Understanding acidity

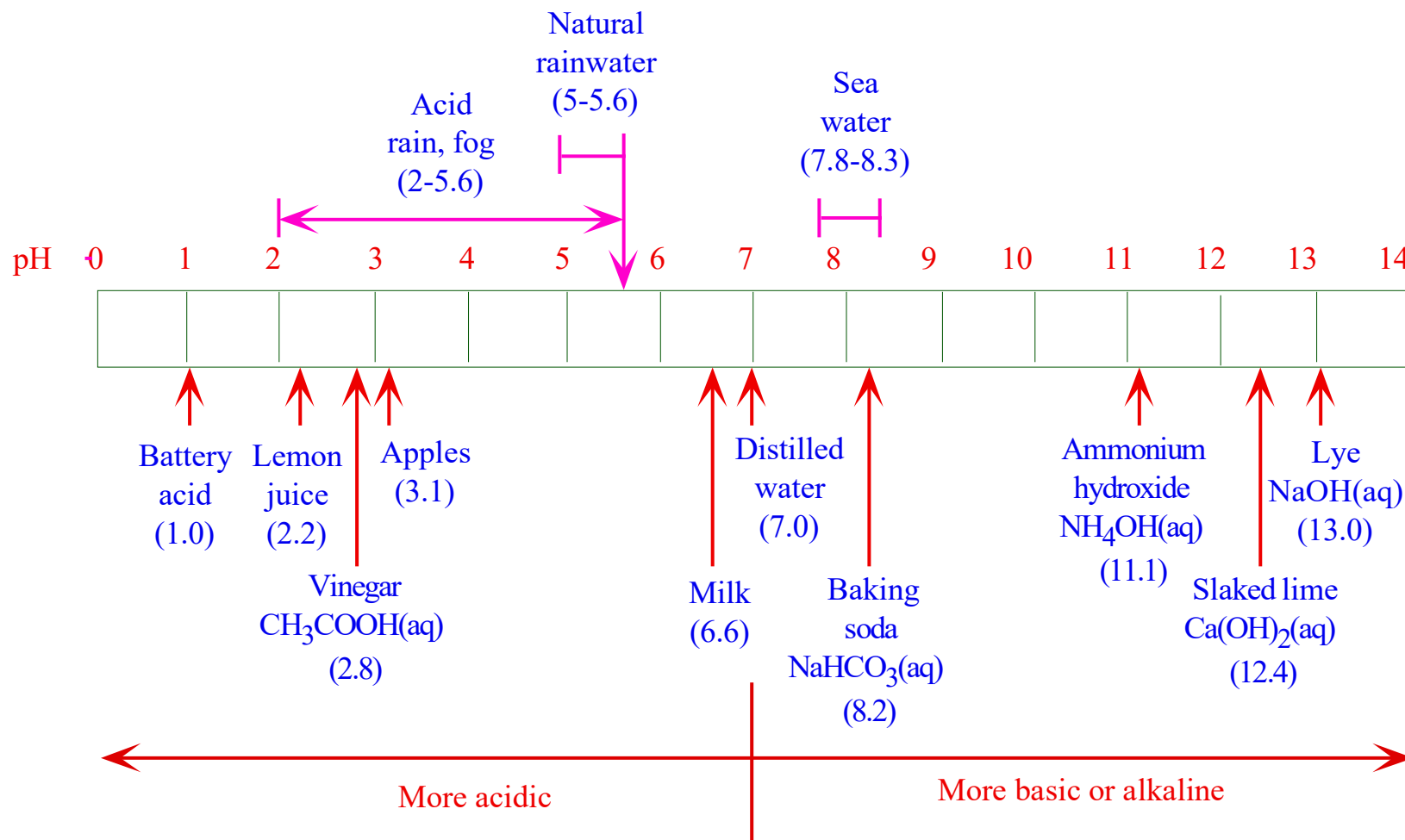
- Acidity ~ the amount of dissociated H^+ ions
 - Characterized by pH value

$$pH = -\log_{10}[H^+]$$

$[H^+]$: molarity (molar or moles/liter)

- Chemical releases $[H^+]$ ions $\rightarrow$ lower pH $\rightarrow$ increase acidity
- Chemical take up $[H^+]$ ions $\rightarrow$ higher pH $\rightarrow$ decrease acidity

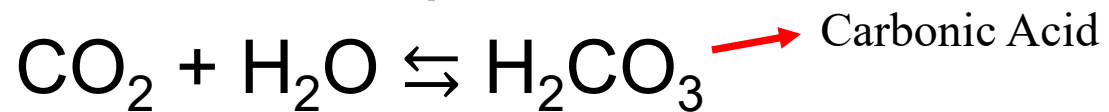
pH Scale



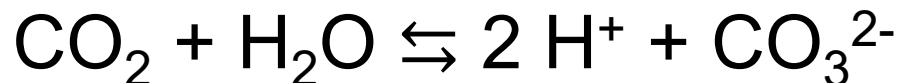
pH scale (Figure 10.3 Jacobson, 2012)

Dissociation of CO₂

- Rain water is slightly acidic naturally (pH ≈ 5.6) due to dissolved CO₂
- Acid rain: rain water with pH < 5



- Overall, we have



Main constituents of acid deposition

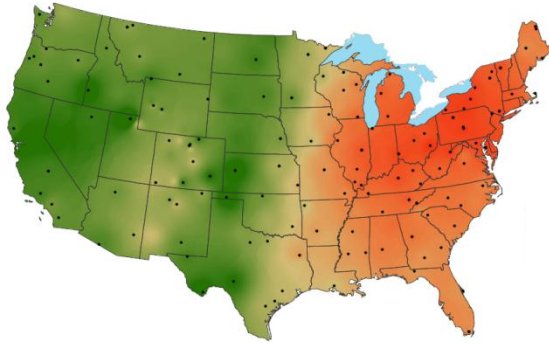
- H_2SO_4
 - Most important in eastern U.S (60 to 70 % of excess acidity)
 - From emitted SO_2
 - 70% of SO_2 emissions are anthropogenic in U.S.
- HNO_3
 - More important in western U.S.
 - Near polluted cities with fog (such as LA)
 - From emitted NO_x
 - 85% of NO_x emissions are anthropogenic in U.S

Main constituents of acid deposition

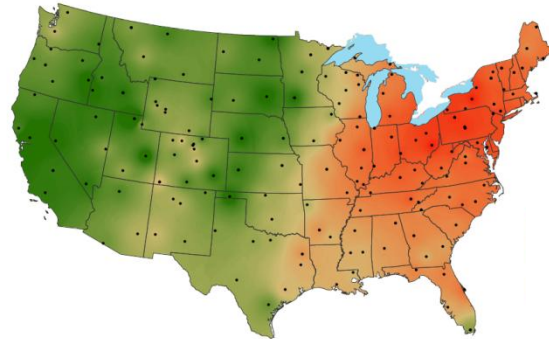
- HCl
 - Wood burning or industrial processing emissions
 - Waste incineration
 - Only about 5% of acidity
- Role of other acids are not as important
 - Formic (HCOOH) & acetic acid (vinegar, CH_3COOH)
- Acid rain control is very successful in US due to increasingly stringent emission controls

pH of rain

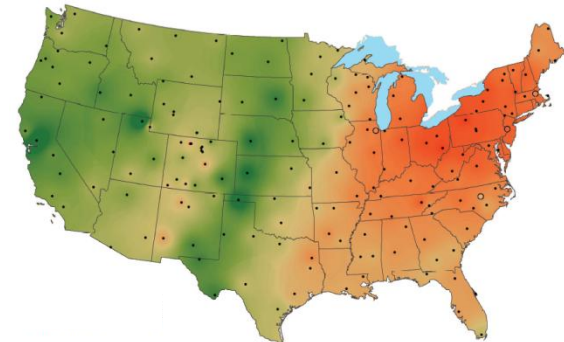
1985



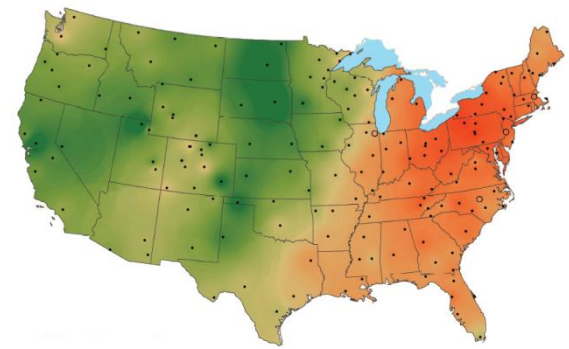
1990



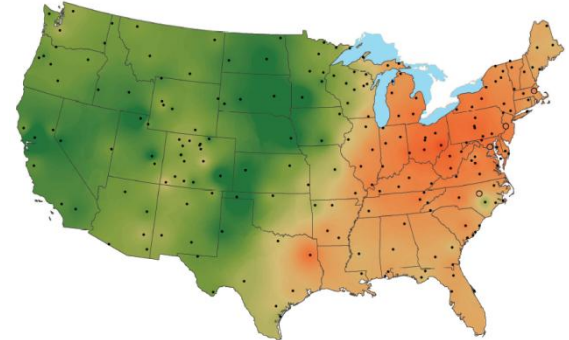
1995



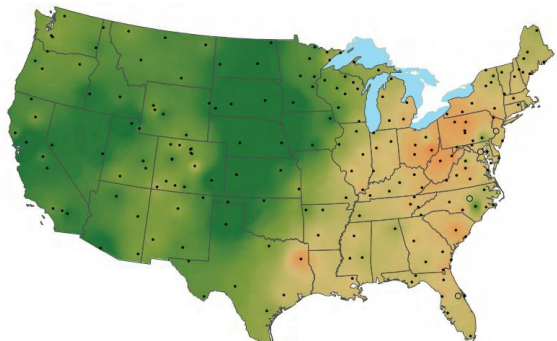
2000



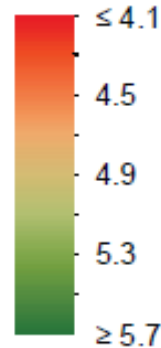
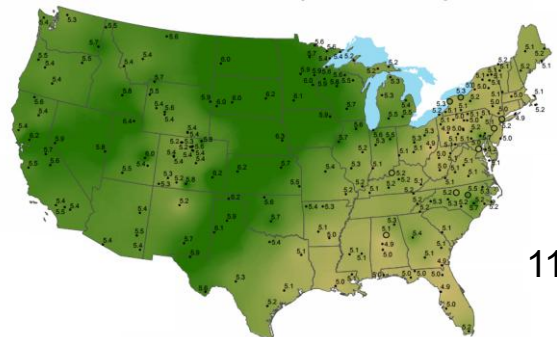
2005



2010



2015



The National Atmospheric Deposition Program (NADP) monitors precipitation chemistry, National Trends Network (NTN), <https://nadp.slh.wisc.edu/>

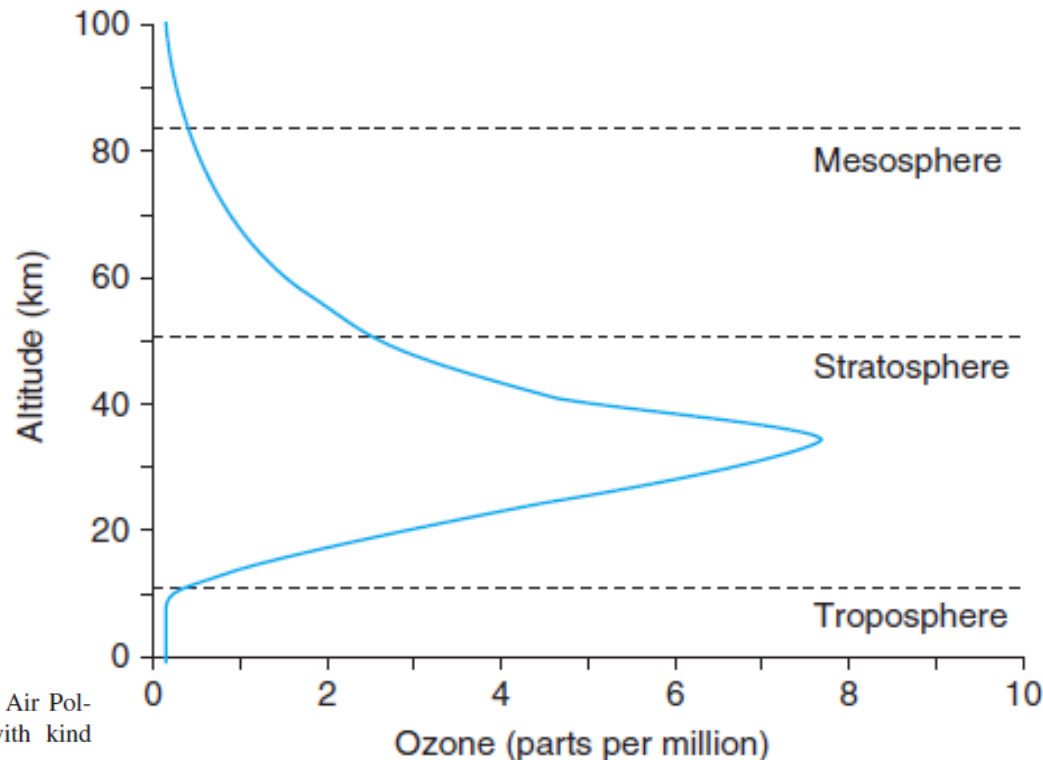
Antarctic Ozone Hole

Introduction

- Ozone presented in both troposphere and stratosphere
- In troposphere → bad ozone
 - Health effect when exposed to
 - Contribute to global warming
- In stratosphere → good ozone
 - Absorbs harmful UV
- 90% of ozone are in stratosphere

Ozone layer structure

- Vertically: peak at 25-32 km altitude
 - Sweet spot with just enough UV and O_2 to form O_3
 - Different chemistry with ground level ozone formation!



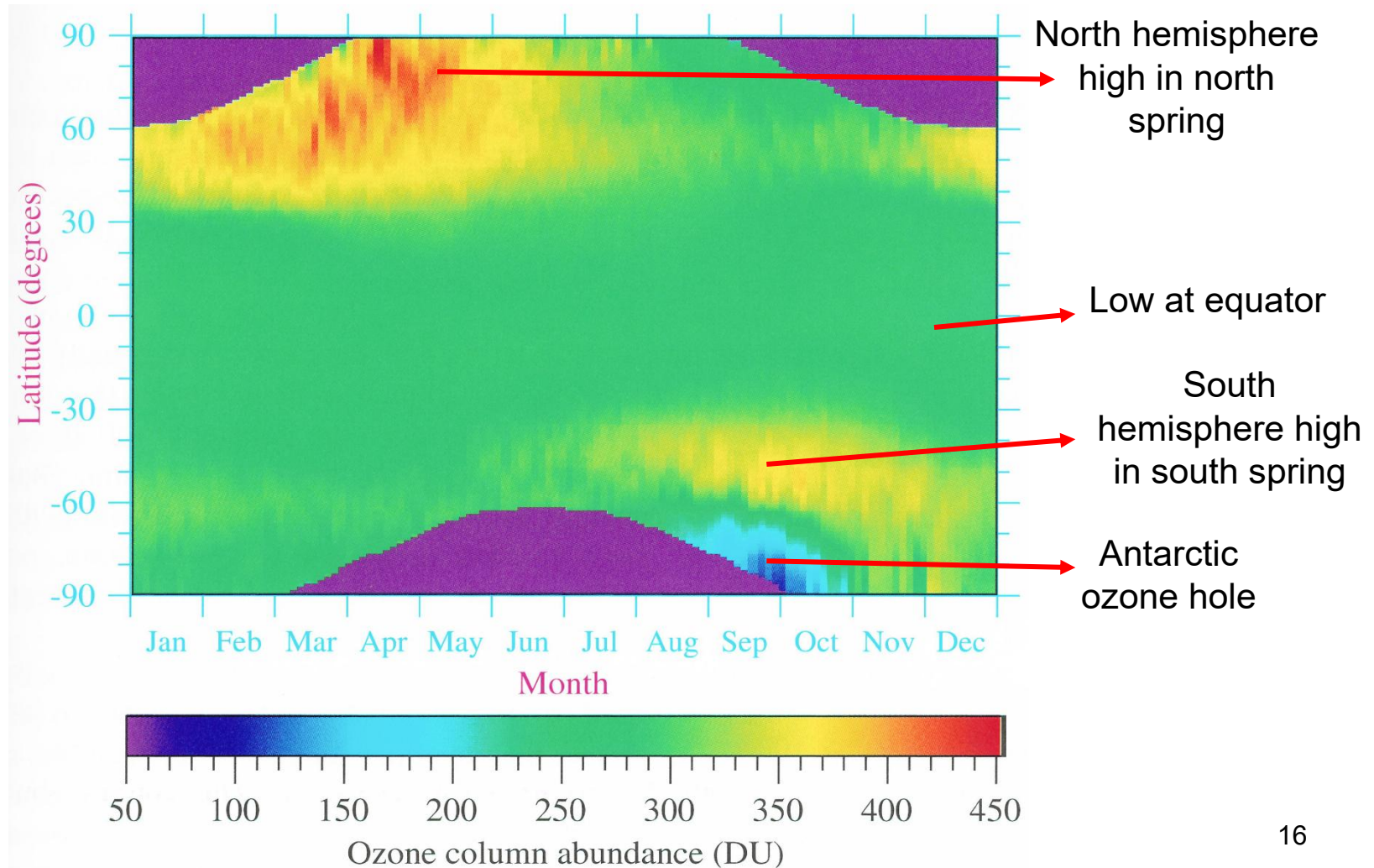
Source: The University of California Air Pollution Health Effects Laboratory, with kind permission.

Ozone layer structure

- Traditionally, ozone concentration expressed in Dubson Unit (DU)
 - Layer of gas that would be 10 μm thick under standard temperature and pressure
 - Globally ozone average concentration ~ 300 DU
- Ozone concentration varies vertically and horizontally

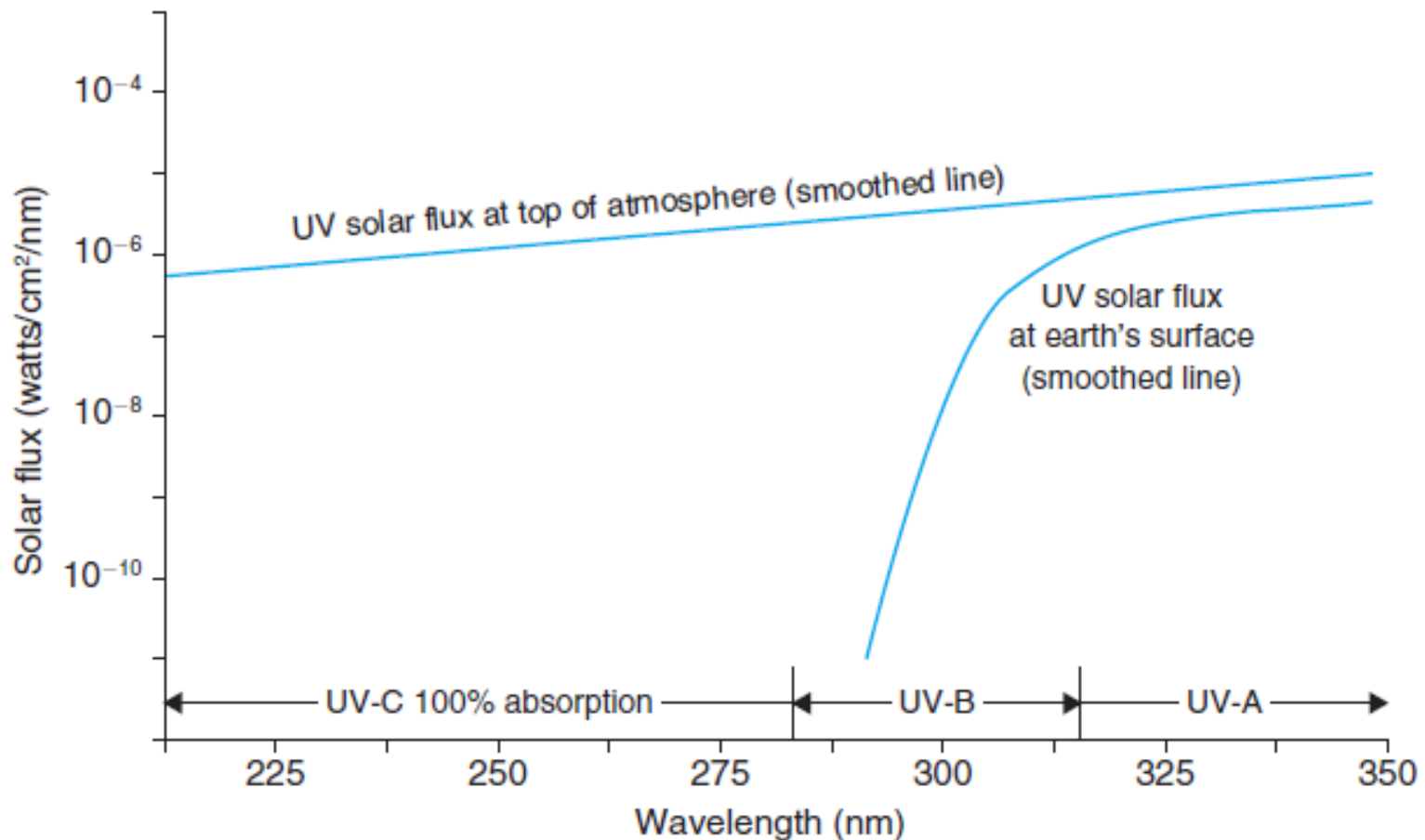
Horizontal ozone layer structure

Ozone concentration variation with time & latitude as measured by NASA TOMS satellite (Jacobson, 2002)



Ozone layer function

- Ozone absorbs harmful UV radiation



Source: The University of California Air Pollution Health Effects Laboratory, with kind permission.

Ozone layer function

- Ozone absorbs UV radiation between 250-345 nm
 - Strongly at 310 nm
 - Weakly between 310-345 nm
 - N_2 , O_2 absorbs far-UV (10-250 nm)
- Allow little UV-C, some UV-B and most UV-A reaches troposphere
- Some say, without ozone layer, no life form as we know will exist

Ozone depletion issues

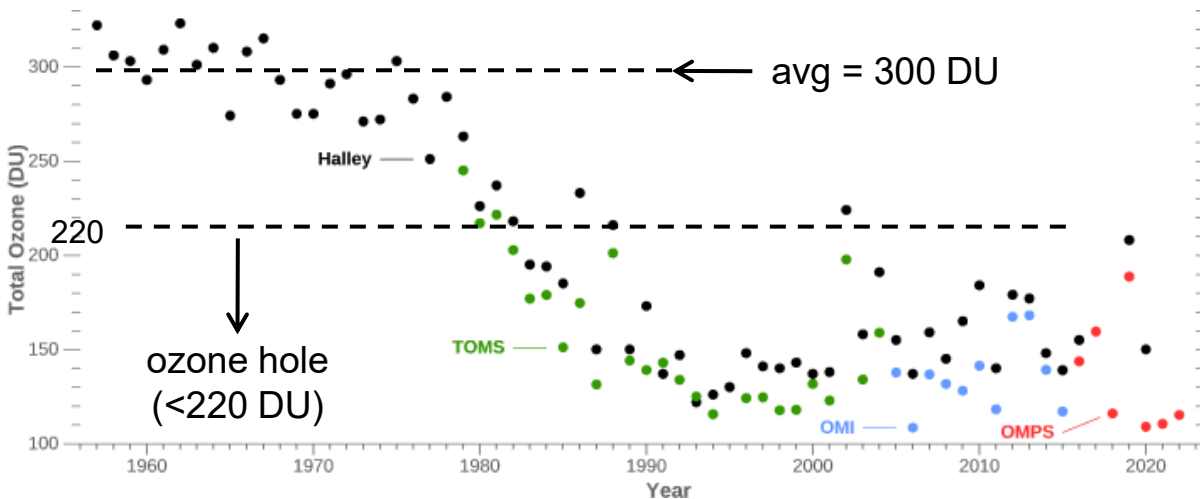
- Largest ozone reduction observed near N/S poles
- Substantial ozone reduction above Antarctic observed at the beginning of southern hemisphere spring
 - Antarctic ozone hole
 - Up to 70% reduction
 - When ozone column abundance < 220 DU: ozone hole
- Ozone reduction in Arctic also observed
 - Less severe (up to 25%)
 - Arctic ozone dent

What is the ozone hole?

The average amount of ozone in the atmosphere is roughly 300 Dobson Units (DU), equivalent to a layer 3 mm thick. Instruments on ground (at Halley) and above Antarctica (Total Ozone Mapping Spectrometer [TOMS] and Ozone Monitoring Instrument [OMI]) measured an acute drop in total atmospheric ozone during October in the early 1980s.

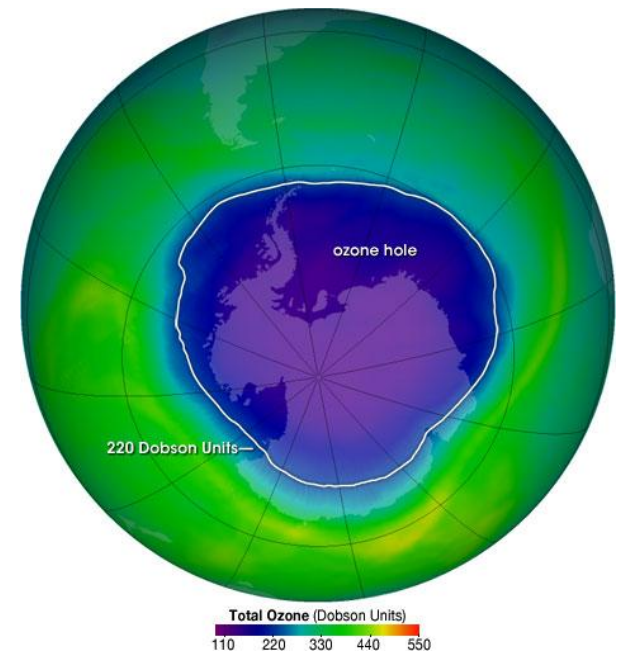
average ozone concentration (1 DU = 0.01 mm thickness)

October data; instruments located on ground at Halley and high above Antarctica (Total Ozone Mapping Spectrometer [TOMS] and Ozone Monitoring Instrument [OMI]).



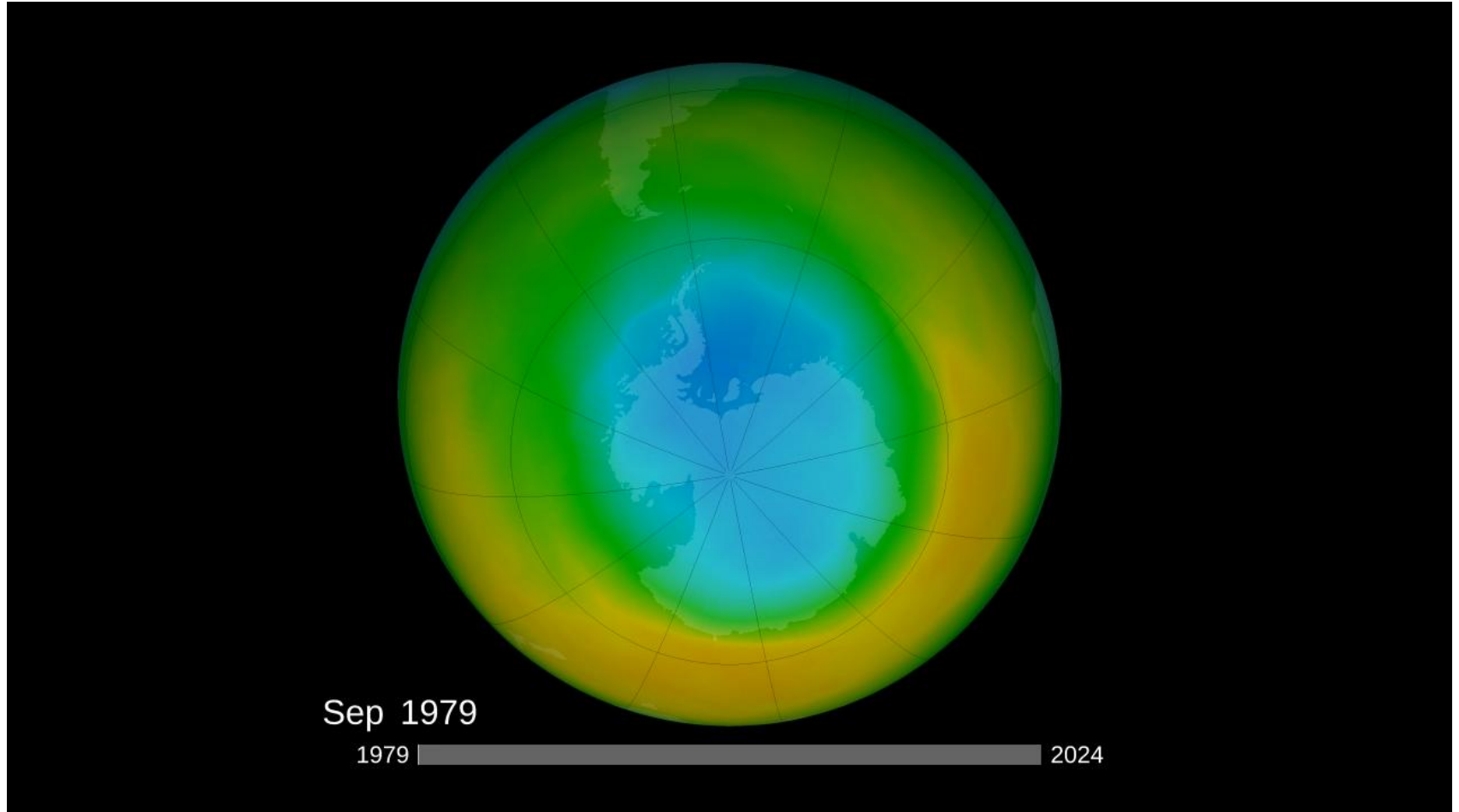
ozone hole area

The ozone “hole” is an area where ozone concentration drops to below 220 DU.



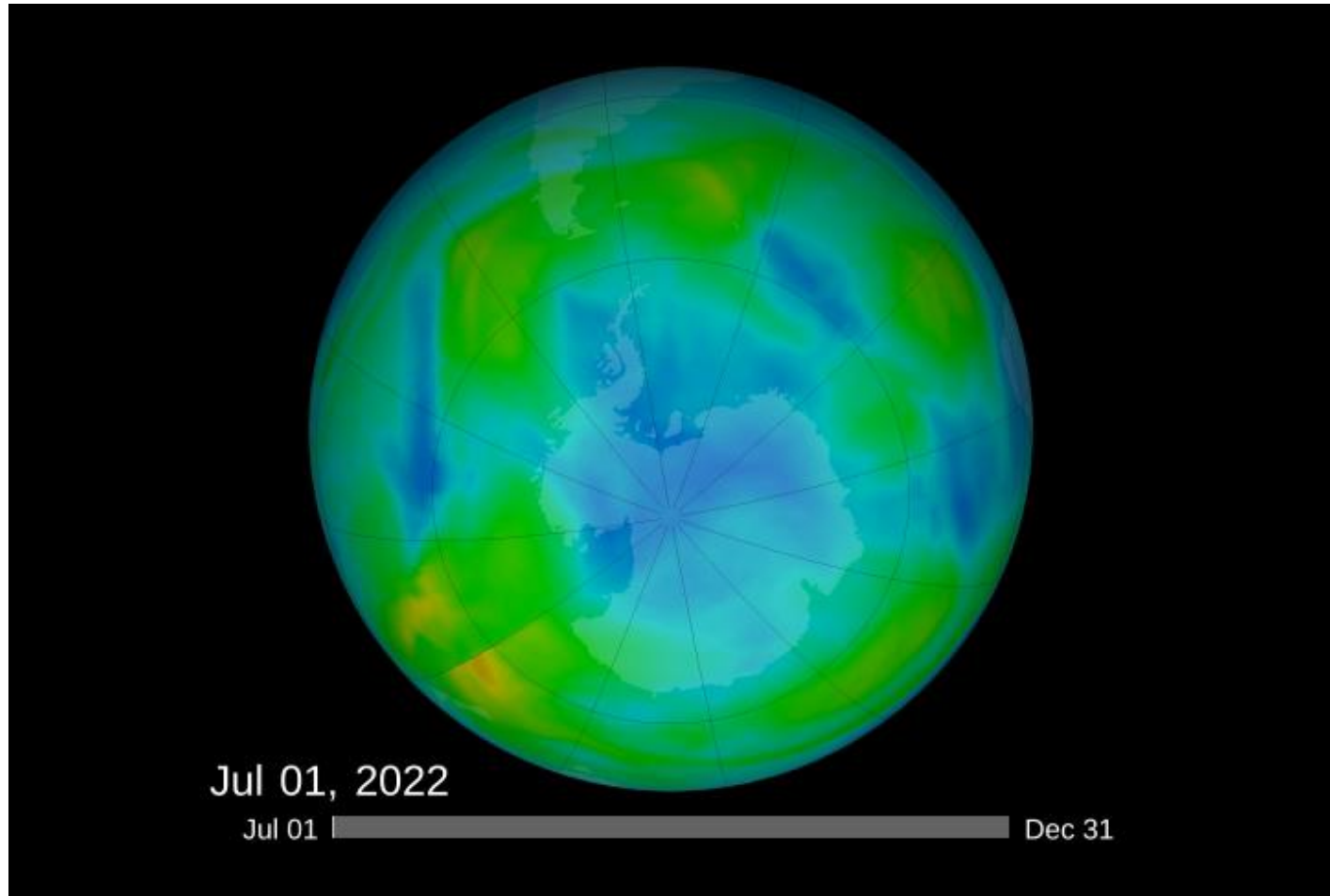
October 4, 2024

Ozone depletion issues



Antarctic ozone hole from 1979 to 2024
(NASA Ozone watch <http://ozonewatch.gsfc.nasa.gov/monthly/SH.html>)

Ozone depletion issues



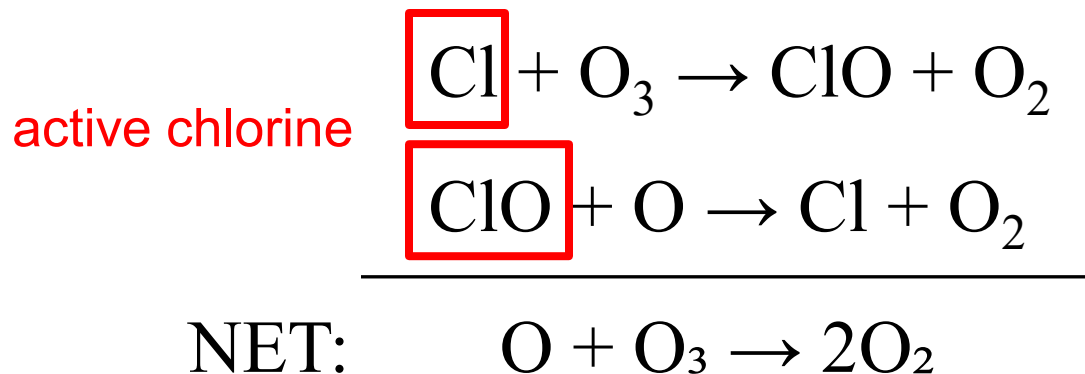
Antarctic ozone hole from 7/1/22 to 12/31/22
(NASA Ozone watch <http://ozonewatch.gsfc.nasa.gov/monthly/SH.html>)

Stratospheric ozone chemistry - CFCs

- Chlorofluorocarbons (CFC)
 - Contains chlorine (Cl) and fluorine (F)
 - Produced synthetically
 - Cheap, nontoxic, nonflammable, nonexplosive, insoluble, relatively inert
 - Used in refrigeration, spray cans, blowing agent
 - Average lifetime ~ 75 years
- There are also other halogenated species: HCFCs, HFCs, bromine containing chemicals...

Stratospheric ozone chemistry - CFCs

- In stratosphere, CFC break down when exposed to far-UV, release ClOx and atomic chlorine
- CFC don't break down in troposphere (why?)
 - Hence climb to stratosphere
- Destruct ozone through the generalized catalytic ozone destruction mechanism



Stratospheric ozone chemistry - CFCs

- -Cl chain length: 10 -1,000 (Jacobson, 2002)
 - One Cl molecule on average destructs 10-1,000 ozone molecules in lower and upper stratosphere, respectively. Aka chain length
 - Bigger chain length number, more powerful ozone destruction
- -Br chain length: 100 - 10,000 (Jacobson, 2002)
 - Slower reaction, more efficient
- The estimated numbers vary

Stratospheric ozone chemistry - CFCs

- Sink of active chlorine (Cl, ClO)
 - Reactions to form reservoir species: HCl and ClONO₂
 - Hydrochloric acid and chlorine nitrate
 - Temporarily store chlorine
 - When sunlight present, can photolyzed back to free chlorine, continue ozone destruction
- Bromine chemistry is similar

Ozone depletion issues

- Ozone reduction near N/S poles generally observed in/near spring time
 - Spring time differs for north/south hemisphere
- One factor: Cl and Br in stratosphere & their chemistry in polar stratospheric clouds (PSC)
 - Recall that active chlorine (Cl, ClO)/Br destroy ozone through generalized catalytic mechanism

Polar stratospheric clouds

Polar Stratospheric Cloud, seen from the NASA DC-8
on 14 January 2003 (photograph by Paul Newman,
GSFC) <http://remus.jpl.nasa.gov/kiruna/a015.htm>



Polar stratospheric clouds

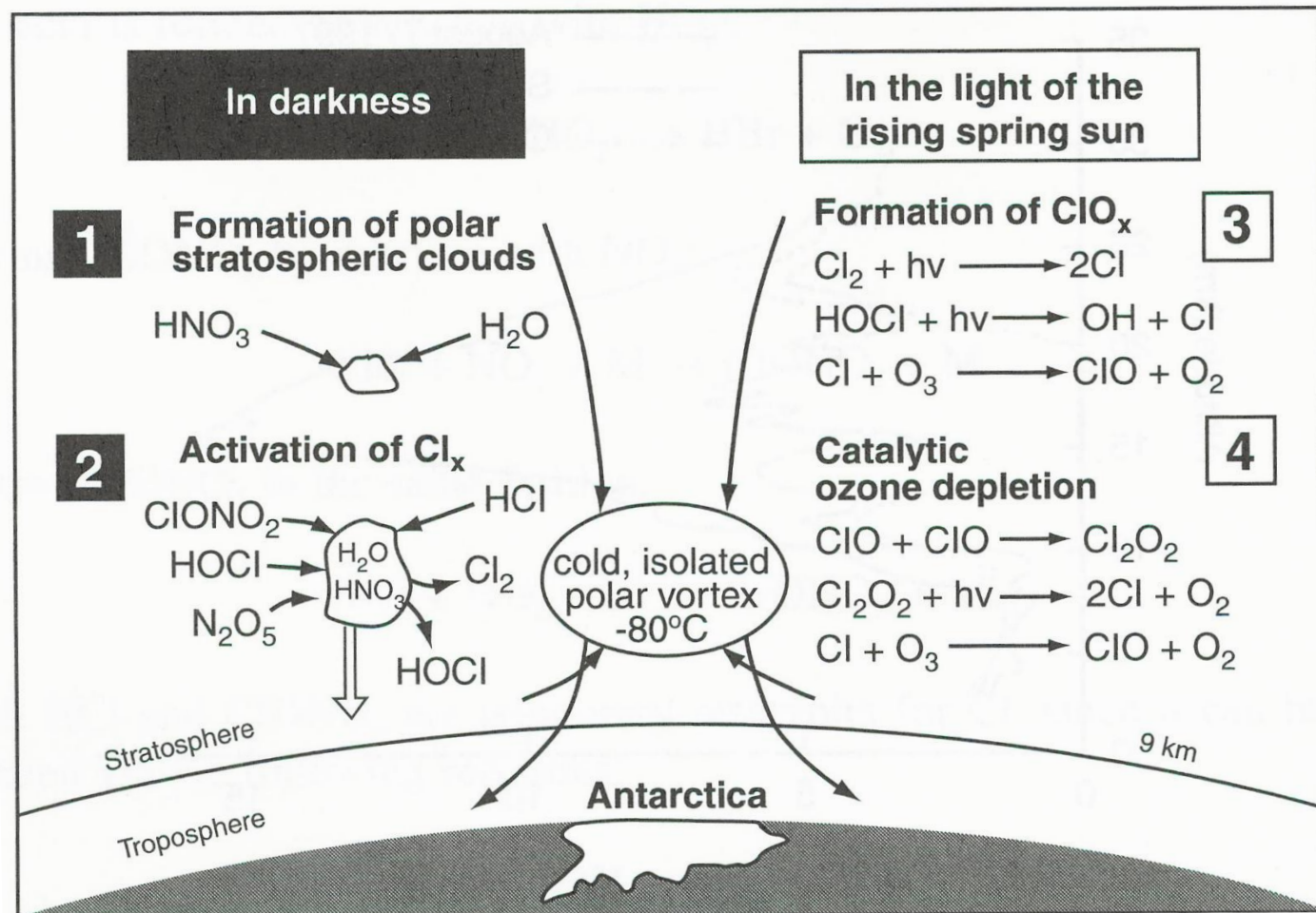


PSC as seen in Scandinavia (source unknown)

Ozone depletion process

- During polar night (before spring)
 - Polar stratospheric clouds (PSC) forms → reactions occur inside PSC → convert inactive Cl/Br (such as reservoir species HCl and ClONO₂) to species that are photo-chemically active (Cl₂, HOCl, ClNO₂ etc)
 - Chlorine activation
- Beginning of polar day (early spring)
 - Photo-chemically active species go through photolysis reaction → release active Cl/Br (Cl, ClO) → catalytically destroy ozone

Ozone depletion process



Ozone depletion process over Antarctica (Figure 4.11 Godish, 2003)

Ozone depletion process

- Why ozone depletion is less severe in north pole?
- Temperature higher in Arctic
 - Less PSC formation, less chlorine activation
 - Higher temperature due to difference in land cover
 - Ocean in Arctic vs land in Antarctic
 - Polar vortex not as strong in arctic due to rough terrain

Stratospheric Ozone Depletion

Websites:

<http://ozonewatch.gsfc.nasa.gov/>

<http://www.ozonelayer.noaa.gov/>

<http://www.youtube.com/watch?v=aKHcHmg4WKY>